QUAID-I-AZAM UNIVERSITY ISLAMABAD

DEPARTMENT OF COMPUTER SCIENCE



DETECTION OF CLICKBAIT THROUGH MACHINE LEARNING

SOFTWARE PROJECT TECHNICAL DOCUMENTATION

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Abstract

Detection of clickbait through Machine Learning

Due to the fast development of social media and information websites, misleading and exaggerated news titles, have become increasingly more commonplace. These misleading information titles goal to seize interest of the reader and maximize the quantity of clicks. However, they'll mislead the reader and create false expectations. In this assignment, the development of a **Clickbait Detection System** based totally on system getting to know may be discussed. The system might be capable to analyze news titles or small information articles to decide whether they're clickbait or valid information.

To increase the **Clickbait Detection System, Natural Language Processing (NLP)** strategies might be hired to procedure the input information, extract essential features, and train a class version to become aware of clickbait, The system might be capable of be given news titles or small news articles as input and offer an immediate prediction with a self assurance level. The machine will also be capable of pick out the vital phrases or terms that inspired the prediction to help the person better recognize the purpose behind the prediction. In addition to this , the system can also be able to maintain a history of the predictions and offer accuracy, precision, and take into account metrices to determine the performance of the type version.

By integrating gadget mastering with the prediction mechanisms, the proposed Clickbait Detection System may be an sensible machine to become aware of misleading news titles. The proposed device could be beneficial for customers to critically examine the virtual content they eat, and the overall recognition of deceptive news titles can be elevated

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Chapter 2

Software Requirement Specification

2.1 Introduction

This document describes the expected software program features, constraints, interfaces and different attributes. In this record we can define instances, sequence diagrams, area model and database layout of the venture.

2.2 Functional Requirements

The functional requirements for the Clickbait Detection Through Machine gaining knowledge of System are as fellow:

1. The system need to permit users to create an account with the aid of presenting valid registration statistics.

2. The device ought to allow registered users to authenticate themselves the usage of stable login credentials.

3. The machine should offer a password recovery mechanism in case customers forget about their credentials.

4. The device ought to permit users to terminate their active session securely.

5. The machine ought to permit users to submit news headlines or article textual content for evaluation.

6. The system have to assist textual content input in English language with varying headline lengths.

7. The system need to preprocess the submitted text by using:

   A. Removing needless symbols and punctuation.

   B. Converting text into a standardized layout.

   C. Handling stop phrases and redundant phrases.

8. The system need to classify the submitted headline or textual content into:

   a. Clickbait

   b. Non-Clickbait

9. The machine ought to calculate a self belief score indicating the chance of the content material being clickbait.

10. The device ought to gift the prediction end result honestly to the person.11. The unit shall provide predictive interpretation records through identifying influential words or terms that make contributions to elegance choice.

12. The unit shall spotlight linguistic styles commonly related to clickbait, which includes:

A. Sensational words

B. Emotional trigger

C. Unclear Terms

D. Hyperbolic expression

13. The machine shall permit customers to upload statistics units in CSV format for analytical purposes.

14. The device have to validate the uploaded records set the usage of checks:

A. Record layout

B. Data consistency

C. Missing or Corrupt Entries

15. The system need to allow legal clients to provoke gaining knowledge of strategies the use of the uploaded datasets.

16. The Gadget shall calculate and display overall performance metrics, which includes:

A. Accuracy

B. Precision

C. Remember

D. F1 score

17. The unit ought to assemble the very last result of every detection:

A. Submitted text

B. Prediction of end end result

C. Confidence assessment

D. Date and time of evaluation

18. The machine must allow users to view formerly analyzed headlines and results.

19. The system must allow customers to go looking and filter detection history based totally on date or prediction kind.

20. The machine must make sure that saved facts stays steady and unaltered throughout sessions.

21. The machine must restriction access to superior analytical features primarily based on person role.

22. The system should ensure that prediction results are generated within an acceptable response time.

23. The system should handle invalid or incomplete input gracefully without system failure.

24. The system should maintain activity logs for analytical and auditing purposes.

Use Case Diagram

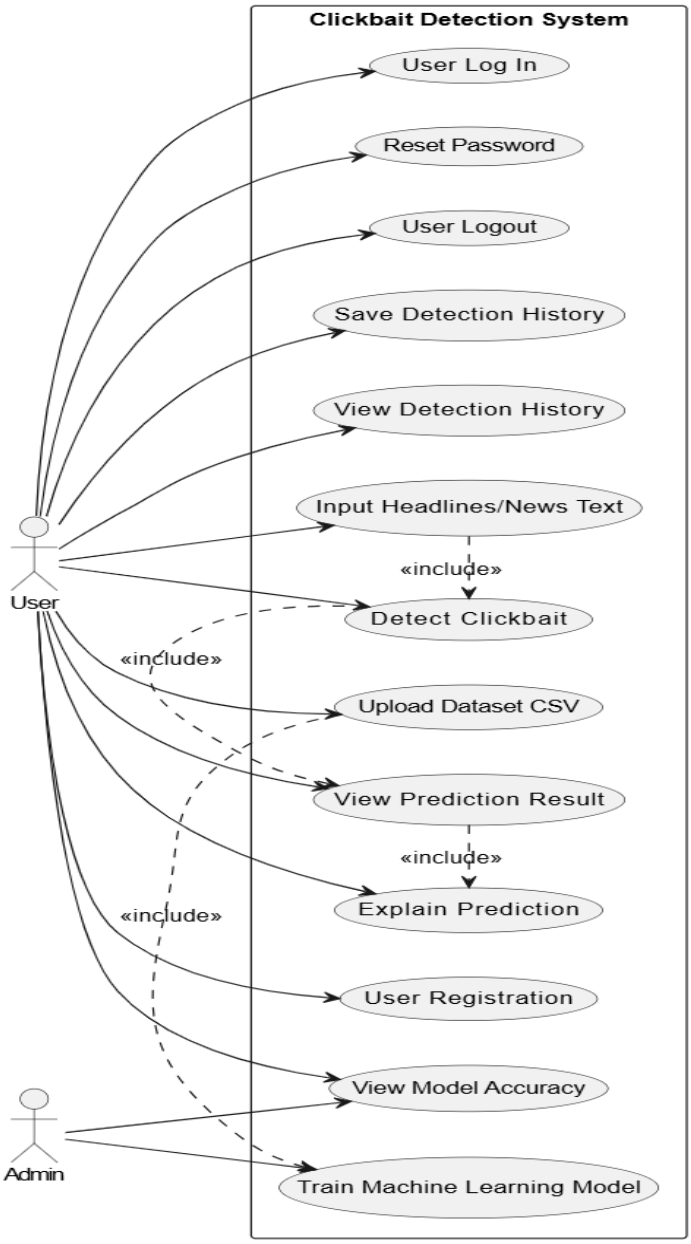


Fig 2 Use Case Diagram

Use Case Description

UC-1 User Registration

USE CASE ID	UC-1
USE CASE NAME	User Registration
Actors	User
Description	It describes how a brand new consumer signs and symptoms up through getting into required statistics.
Preconditions	1. User has access to the Clickbait Detection System. 2. User is not already registered in the system.
Postconditions	A new user account is successfully created. User information is securely stored in the SYSTEM.
Normal Flow	1. User opens the registration interface. 2. User enters required details along with name, electronic mail, and password. 3. User submits the registration shape. 4. The gadget validates the entered information. 5. The device creates a brand new user account. 6. The device displays a a success registration confirmation.
Alternative Flow	**3a.** If any required field is missing, the system prompts the user to complete the missing information. **3b.** If the entered email already exists, the system notifies the user to use a different email address
Exception	1. Invalid input format. 2. System failure during the registration process.
Special Requirements	1. User credentials need to be saved in encrypted shape. 2. Email addresses need to be unique across the system. 3. The registration manner have to whole within perfect reaction time.
Frequency of Use	Low
Technology and Data variation List	1. Registration is performed thru a web-based totally interface. 2. User input records can also vary in period and layout. 3. The device should assist multiple customers registering concurrently.

Table 1 Use Case Description UC-1 User Registration

UC-2 User Login

USE CASE ID	UC-2
USE CASE NAME	User Login
Actors	User

Description	It describes how a consumer signs in the usage of a legitimate username and password.
Preconditions	1. User must be registered. 2. User must have valid login credentials
Postconditions	1. User is efficiently authenticated. 2. User is redirected to the system dashboard.
Normal Flow	1. User opens login page. 2. User enters email and password. 3. User submits login request. 4. System verifies credentials. 5. System grants access.
Alternative Flow	**3a.** If credentials are incorrect, SYSTEM displays an error message. **3b.** If fields are empty, SYSTEM prompts user to complete them.
Exception	System authentication failure.
Special Requirements	Secure authentication mechanism must be used.
Frequency of Use	High
Technology and Data variation List	Web-based login, variable credential formats.

Table 2 Use Case Description UC-2 User Login

UC-3 Reset Password

USE CASE ID	UC-3
USE CASE NAME	Reset Password
Actors	User
Description	It shows the technique of password recuperation for registered users.
Preconditions	User must be registered.
Postconditions	Password is successfully updated.
Normal Flow	1. User selects reset password choice. 2. User enters registered e mail. 3. System verifies electronic mail. 4. User units new password. 5. System updates password.
Alternative Flow	**2a.** Email not found → SYSTEM shows error.
Exception	SYSTEM error during password update.
Special Requirements	Password reset must follow security rules.
Frequency of Use	Low

Technology and Data variation List	Email-based verification, varied password formats.

Table 3 Use Case Description UC-3 Reset Password

UC-4 User Logout

USE CASE ID	UC-4
USE CASE NAME	User Logout
Actors	User
Description	It describes how a person symptoms out of the system.
Preconditions	User must be logged in.
Postconditions	User session is terminated.
Normal Flow	1. User selects logout choice. 2. System ends consultation. 3. User is redirected to login web page.
Alternative Flow	N/A
Exception	Session termination failure.
Special Requirements	Session termination failure.
Frequency of Use	Medium.
Technology and Data variation List	Session-based authentication handling.

Table 4 Use Case Description UC-4 User Logout

UC-5 Input Headlines / News Text

USE CASE ID	UC-5
USE CASE NAME	Input Headlines / News Text
Actors	User
Description	This use case permits customers to analyze headlines or news textual content.
Preconditions	User must be logged in.
Postconditions	Text is submitted for processing.
Normal Flow	1. User opens text input interface. 2. User enters headline or news text. 3. User submits text.
Alternative Flow	**2a.** Empty input → system prompts user.
Exception	Invalid text format.
Special Requirements	Input length limits must be enforced.
Frequency of Use	High

Technology and Data variation List	Text length and language variation.

Table 5 Use Case Description UC-5 Input Headlines / News Text

UC-6 Upload Dataset

USE CASE ID	UC-6
USE CASE NAME	Upload Dataset
Actors	User
Description	This use case explains how customers add statistics to improve the model.
Preconditions	User must be logged in.
Postconditions	Dataset is stored in the system.
Normal Flow	1. User selects add dataset option. 2. User uploads CSV report. 3. System validates report layout. 4. Dataset is saved.
Alternative Flow	**2a.** Invalid file → system rejects upload.
Exception	File processing error.
Special Requirements	Only CSV format allowed.
Frequency of Use	Low
Technology and Data variation List	Only CSV format allowed.

Table 6 Use Case Description UC-6 Upload Dataset